# Activity A5 - Design Life-Cycle Operational Controls

Course: Fundamentals of ISO 14001 Environmental Management System (EMS) (TGS-2023037592) Ability: A5 Objective: Evaluate significant aspects and design controls across activities and externally provided processes, products and services. Mode: Team analysis with individual evidence submission Suggested time: 60 minutes

## Scenario

MEP controls the printing line but has weak requirements for paper suppliers, cleaning contractors, waste collectors and transport providers. Instructions describe tasks but not environmental acceptance criteria.

Meridian EcoPack Pte Ltd (MEP) is a fictional Singapore SME with 120 employees producing printed corrugated and fibre-based packaging for regional food, electronics and healthcare customers. Two key customers require evidence of an ISO 14001:2026-aligned EMS and stronger value-chain environmental controls before contract renewal.

## Task and expected output

Control ink, paper, cleaning, maintenance, waste handling and supplier interfaces using the hierarchy of control and life-cycle perspective.

Produce: An operational control plan, supplier requirement matrix and control-verification schedule.

## Before you begin

- Open a working copy of the Excel workbook; do not overwrite the supplied evidence.
- Allocate a facilitator, evidence recorder, checker and presenter.
- Read every evidence card before making a compliance or significance conclusion.
- If the activity concerns an incident, this is a tabletop exercise: do not handle chemicals or simulate unsafe physical actions.

## Decision questions

1. Where can MEP control an activity directly and where can it only influence outcomes?
2. Which controls prevent or reduce impact at source before relying on administrative checks?
3. What acceptance evidence is proportionate for externally provided processes, products and services?

## Detailed procedure

### Step 1 - Map the value-chain stages from material selection through manufacture, delivery, use and end-of-life influence

Map the value-chain stages from material selection through manufacture, delivery, use and end-of-life influence.

Record the evidence used, the decision made, the responsible owner and any uncertainty in the workbook.

### Step 2 - Link significant aspects and compliance obligations to each relevant stage and provider

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### Step 3 - Select controls using elimination/reduction at source, engineering, administrative and response layers

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### Step 4 - Define operating criteria, responsibility, evidence, monitoring method and change trigger for each control

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### Step 5 - Write supplier/contractor environmental requirements and proportionate approval or performance evidence

Write supplier/contractor environmental requirements and proportionate approval or performance evidence.

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### Step 6 - Evaluate three current controls for implementation and effectiveness; record improvements and owners

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Record the evidence used, the decision made, the responsible owner and any uncertainty in the workbook.

## Evidence to submit

- Completed Activity-A5-Workbook.xlsx.
- One evidence-backed team decision and one documented peer challenge.
- Any assumption or missing source clearly labelled for follow-up.

## Acceptance checklist

[ ] Controls address the actual cause of impact, include operating criteria and evidence, distinguish control from influence, and extend proportionately across relevant value-chain interfaces.

[ ] Claims are traceable to supplied or cited evidence.

[ ] Assumptions, uncertainty and missing evidence are labelled.

[ ] Owners, due dates and next decisions are included where applicable.

[ ] Legal/compliance interpretations needing competent advice are escalated rather than guessed.

[ ] A peer challenge and response are recorded.

## Troubleshooting

- The team cannot agree on a score: return to the criterion anchors and record each evidence-based assumption before averaging or escalating.
- Evidence is missing: do not invent it; record an evidence gap, risk implication, owner and retrieval action.
- A legal requirement is unclear: record the current authoritative source and escalate to a competent adviser or authority channel.
- A conclusion feels obvious: test an alternative explanation and state what evidence would falsify the current view.

## Facilitator debrief

Ask each team to defend one priority, control or conclusion using objective evidence. Probe what would change the decision, then connect the work to PDCA and the relevant EMS process rather than clause memorisation.

## Files in this activity pack

- README.md and README.pdf - detailed learner procedure.
- Activity-A5-Scenario-and-Checklist.pdf - printable scenario and completion checklist.
- Activity-A5-Workbook.xlsx - editable analysis and submission workbook.

## Source and usage note

This fictional training activity paraphrases ISO 14001 concepts and is not legal advice or a reproduction of the standard. Use the organisation's licensed ISO 14001:2026 copy and current authoritative Singapore sources for formal interpretation.